

Technical Document #739: Retrieval Augmented Generation - Comprehensive Overview

Technical Document #739: Retrieval Augmented Generation - Comprehensive Overview

Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Evaluation metrics for RAG include faithfulness, answer relevance, and context recall. They measure the quality of the retrieval-generation pipeline.

Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

Key Takeaways:

- Retrieval-Augmented Generation (RAG) combines information retrieval with text generation.
- Semantic search finds documents based on meaning rather than exact keywords.
- Hybrid search combines keyword-based and semantic search.